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Lecture Activity 2

IT325 Machine Learning | EDA Planning for the Performance Innovative Task

Type	Lecture Activity
Topic Focus	Introduction to exploratory data analysis for text classification
Time Allotment	45 to 60 minutes
PIT Connection	Students inspect and plan the EDA phase of their approved dataset
Mode	Individual classroom activity

Learning Outcomes

1. Identify the text column and target variable in an approved PIT dataset.
2. Describe the size, structure, and basic quality of the dataset.
3. Plan suitable EDA steps before model training begins.

Activity

In this activity, students will examine their approved text classification dataset and prepare an EDA plan. The goal is to understand the problem, the available variables, and the quality of the data before any preprocessing or model training is performed.

Materials Needed

1. Approved dataset link or CSV file
2. Notebook, worksheet, or digital document for responses
3. Laptop for quick dataset inspection if available

Classroom Procedure

1. The instructor gives a short discussion on why EDA is necessary in machine learning projects.
2. Students review the PIT requirements and confirm that their dataset matches the approved topic and format.
3. Students inspect the dataset and answer the guide questions individually.
4. Selected students share their observations and proposed EDA steps.

Expected Output

A one page EDA planning sheet or short written response that describes the dataset, the target variable, possible data quality issues, and the EDA tasks to be done in the laboratory.

Questions

1. What is the title or source of your dataset?

Student Depression Dataset, sourced from Kaggle (https://www.kaggle.com/datasets/hopesb/student-depression-dataset).

2. What real world problem does your dataset help solve?

This dataset helps identify which students are at risk of depression based on academic, lifestyle, and socioeconomic factors. Early detection of depression in students can help schools and counselors provide timely support and mental health interventions.

3. Which column contains the text data?

This dataset does not contain a single raw text column. Instead, it uses a combination of structured/categorical features such as Sleep Duration, Dietary Habits, Have you ever had suicidal thoughts?, and Degree as text-based inputs, along with numeric features like Academic Pressure, CGPA, Financial Stress, and Work/Study Hours.

4. What is your target variable?

The target variable is the Depression column, which is a binary label where **1** means the student is depressed and **0** means the student is not depressed.

5. How many rows and columns are in the dataset?

The dataset contains 27,901 rows and 18 columns.

6. Are there missing values, duplicates, or columns that may not be useful?

Missing Values: Only 3 missing values were found in the Financial Stress column. These can be handled by filling them with the column median or by dropping the affected rows.=

Duplicates: No duplicate rows were detected in the dataset.

Possibly Irrelevant Columns: The id column is just a row identifier and does not provide useful information for model training, so it can be dropped. The City and Profession columns may also have limited predictive value depending on the model used.

7. What EDA steps will you perform before training models?

1. Load the dataset and display the first few rows to understand its structure.
2. Check the dataset shape, column names, and data types.
3. Identify and handle missing values (fill or drop the 3 missing Financial Stress values).
4. Check for and confirm there are no duplicate rows.
5. Analyze the class distribution of the target variable (Depression) to detect any imbalance.
6. Create visualizations: class distribution bar and pie chart, age distribution histogram and boxplot, academic pressure vs. depression count plot, CGPA distribution histogram, sleep duration and dietary habits grouped bar charts, correlation heatmap of numeric features, and financial stress vs. depression count plot.
7. Write observations and interpretation based on the EDA findings.

Screenshot of your code

Exploratory Data Analysis: Student Depression Dataset

Lab Activity – Text Classification EDA

John Lemar L. Gonzales - BSIT3-R15

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1. Setup and Imports

[ ]:
```
# First, we import the tools we need
import pandas as pd                # for working with tables/data
import numpy as np                 # for math stuff
import matplotlib.pyplot as plt    # for making charts
import seaborn as sns              # for nicer-looking charts
import warnings
warnings.filterwarnings('ignore') # hide unnecessary warnings

# This just makes our charts look cleaner
sns.set_theme(style="whitegrid")
plt.rcParams["figure.dpi"] = 120

print("All libraries loaded!")
```

2. Dataset Overview

```
df = pd.read_csv(r"C:\Lemar\GithubRepositories\student-depression-detector\notebooks\Student Depression Dataset.csv")

print("First 5 rows of the dataset:")
df.head()
```

```
First 5 rows of the dataset:
```

	id	Gender	Age	City	Profession	Academic Pressure	Work Pressure	CGPA	Study Satisfaction	Job Satisfaction	Sleep Duration	Dietary Habits	Degree	Have you ever had suicidal thoughts ?	Work/Study Hours	Financial Stress
0	2	Male	33.0	Visakhapatnam	Student	5.0	0.0	8.97	2.0	0.0	5-6 hours	Healthy	B.Pharm	Yes	3.0	1.0
1	8	Female	24.0	Bangalore	Student	2.0	0.0	5.90	5.0	0.0	5-6 hours	Moderate	BSc	No	3.0	2.0
2	26	Male	31.0	Srinagar	Student	3.0	0.0	7.03	5.0	0.0	Less than 5 hours	Healthy	BA	No	9.0	1.0
3	30	Female	28.0	Varanasi	Student	3.0	0.0	5.59	2.0	0.0	7-8 hours	Moderate	BCA	Yes	4.0	5.0
4	32	Female	25.0	Jaipur	Student	4.0	0.0	8.13	3.0	0.0	5-6 hours	Moderate	M.Tech	Yes	1.0	1.0

```
print("Shape (rows, columns):", df.shape)

print("\nColumn Names:")
for col in df.columns:
    print(" -", col)
```

```
Shape (rows, columns): (27901, 18)

Column Names:
 - id
 - Gender
 - Age
 - City
 - Profession
 - Academic Pressure
 - Work Pressure
 - CGPA
 - Study Satisfaction
 - Job Satisfaction
 - Sleep Duration
 - Dietary Habits
 - Degree
 - Have you ever had suicidal thoughts ?
 - Work/Study Hours
 - Financial Stress
 - Family History of Mental Illness
 - Depression
```

```python
# Check data type
print("Data Types of Each Column:")
print(df.dtypes)
```

```
Data Types of Each Column:
id                                      int64
Gender                                    str
Age                                   float64
City                                      str
Profession                                str
Academic Pressure                     float64
Work Pressure                         float64
CGPA                                  float64
Study Satisfaction                    float64
Job Satisfaction                      float64
Sleep Duration                            str
Dietary Habits                            str
Degree                                    str
Have you ever had suicidal thoughts ?     str
Work/Study Hours                      float64
Financial Stress                      float64
Family History of Mental Illness          str
Depression                              int64
dtype: object
```

3. Data Quality Checks

```python
# Check for missing (empty) values in each column
print("Missing values per column:")
print(df.isnull().sum())
```

```
Missing values per column:
id                                      0
Gender                                  0
Age                                     0
City                                    0
Profession                              0
Academic Pressure                       0
Work Pressure                           0
CGPA                                    0
Study Satisfaction                      0
Job Satisfaction                        0
Sleep Duration                          0
Dietary Habits                          0
Degree                                  0
Have you ever had suicidal thoughts ?   0
Work/Study Hours                        0
Financial Stress                        3
Family History of Mental Illness        0
Depression                              0
dtype: int64
```

```python
# Check for duplicate rows
duplicates = df.duplicated().sum()
print("Number of duplicate rows:", duplicates)
```

```
Number of duplicate rows: 0
```

4. Class Distribution (Target Variable)

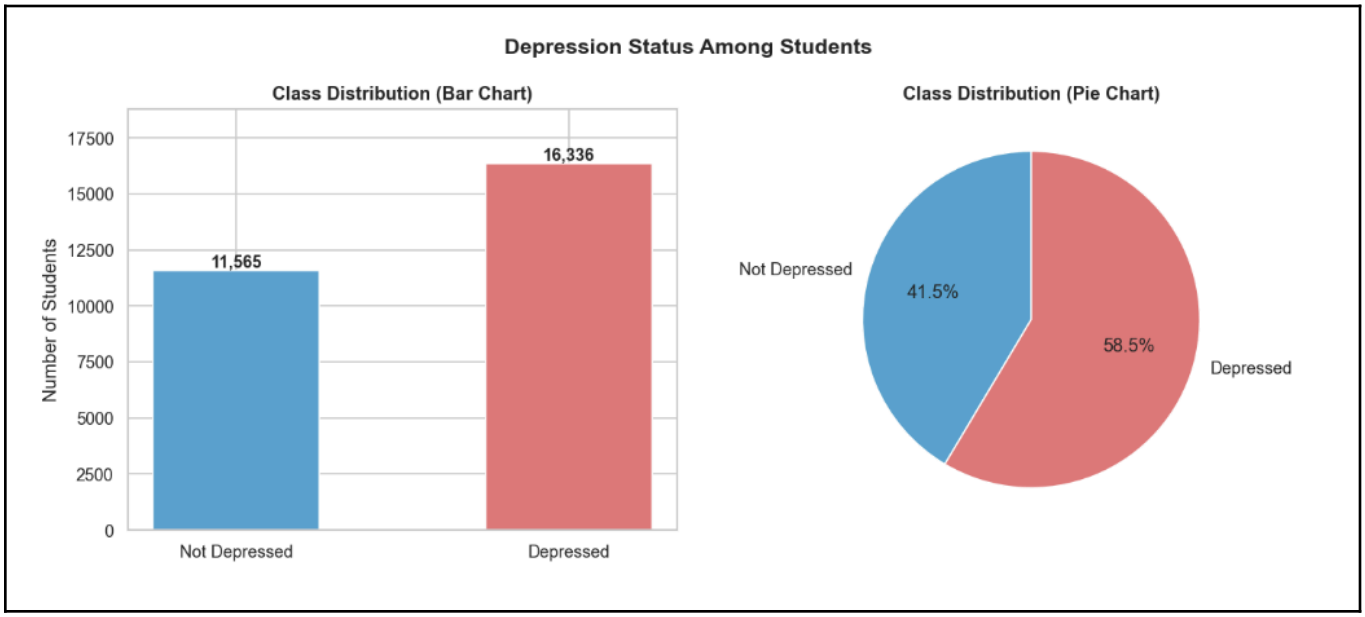
```
# Count how many students are depressed (1) vs not depressed (0)
print("Depression column value counts:")
print(df["Depression"].value_counts())

print()

# Show percentage also
total = len(df)
for val, count in df["Depression"].value_counts().items():
    label = "Depressed" if val == 1 else "Not Depressed"
    print(f"{label}: {count} students ({count/total*100:.1f}%)")
```

```
Depression column value counts:
Depression
1    16336
0    11565
Name: count, dtype: int64

Depressed: 16336 students (58.5%)
Not Depressed: 11565 students (41.5%)
```

5. EDA Visualizations

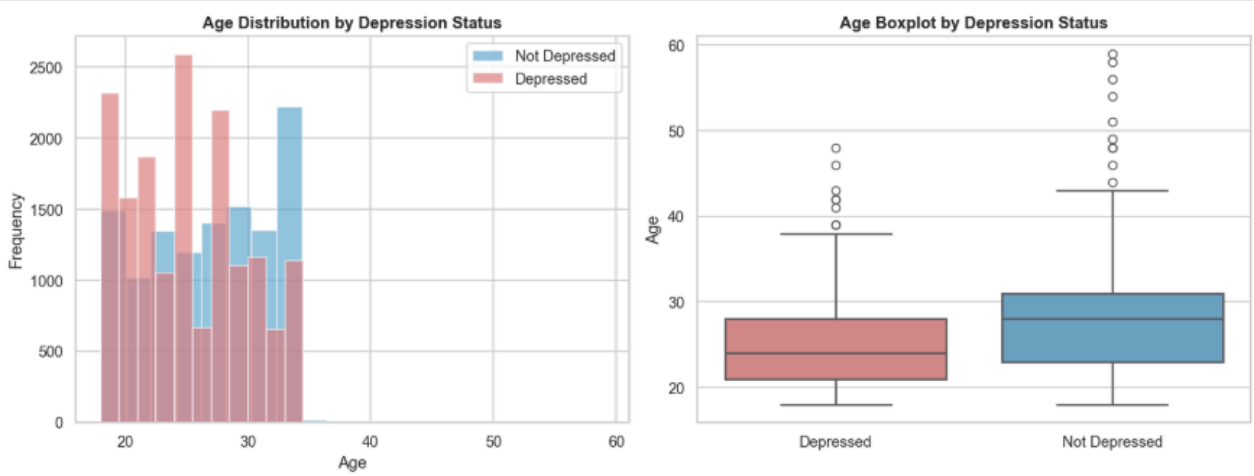
5.1 Target Class Distribution

```
# We want to see how many students are depressed vs not depressed
labels = ["Not Depressed", "Depressed"]
counts = df["Depression"].value_counts().sort_index().values
colors = ["#5BA4CF", "#E07B7B"]

# Create two charts side by side
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# --- Left chart: Bar chart ---
bars = axes[0].bar(labels, counts, color=colors, edgecolor="white", width=0.5)
axes[0].set_title("Class Distribution (Bar Chart)", fontweight="bold")
axes[0].set_ylabel("Number of Students")
axes[0].set_ylim(0, max(counts) * 1.15)

# Add the number on top of each bar
for bar, count in zip(bars, counts):
    axes[0].text(bar.get_x() + bar.get_width()/2,
                 bar.get_height() + 150,
                 f"{count:,}", ha="center", fontsize=11, fontweight="bold")

# --- Right chart: Pie chart ---
axes[1].pie(counts, labels=labels, autopct="%1.1f%%",
            colors=colors, startangle=90)
axes[1].set_title("Class Distribution (Pie Chart)", fontweight="bold")

plt.suptitle("Depression Status Among Students", fontsize=14, fontweight="bold")
plt.tight_layout()
plt.savefig("viz1_class_distribution.png", bbox_inches="tight")
plt.show()
```


Depression Status Among Students
Class Distribution (Bar Chart)
Number of Students
17500
15000
12500
10000
7500
5000
2500
0
11,565
16,336
Not Depressed
Depressed
Class Distribution (Pie Chart)
Not Depressed
41.5%
58.5%
Depressed

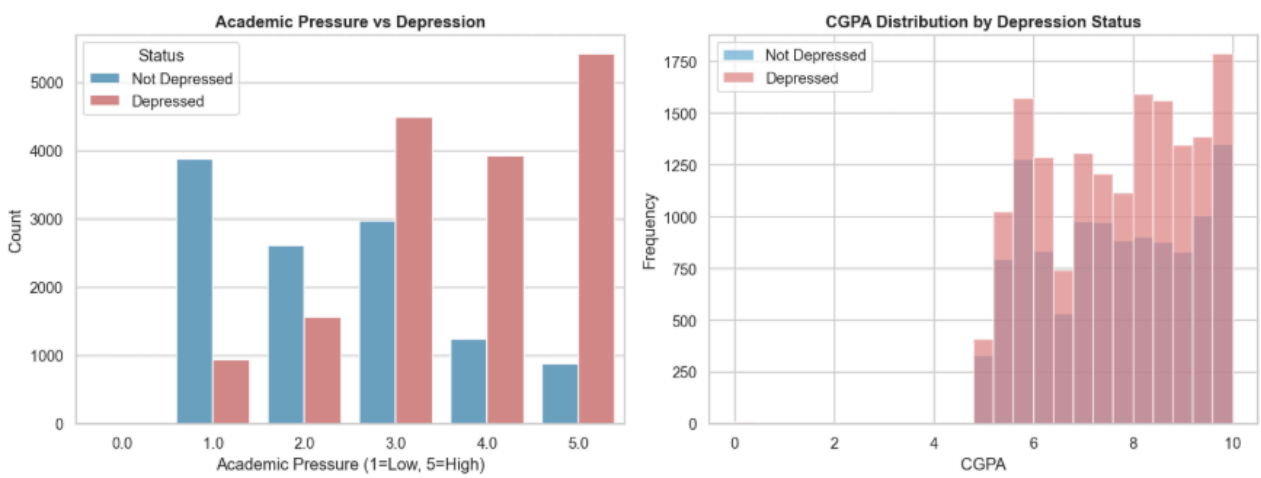
5.2 Age Distribution by Depression Status

```python
# Add a label column so charts are easier to read
df["Depression_Label"] = df["Depression"].map({0: "Not Depressed", 1: "Depressed"})

fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# --- Left: Histogram to see age spread ---
for val, label, color in [(0, "Not Depressed", "#5BA4CF"), (1, "Depressed", "#E07B7B")]:
    subset = df[df["Depression"] == val]["Age"]
    axes[0].hist(subset, bins=20, alpha=0.65, label=label, color=color, edgecolor="white")

axes[0].set_title("Age Distribution by Depression Status", fontweight="bold")
axes[0].set_xlabel("Age")
axes[0].set_ylabel("Frequency")
axes[0].legend()

# --- Right: Boxplot to compare age ranges ---
sns.boxplot(data=df, x="Depression_Label", y="Age",
            palette={"Not Depressed": "#5BA4CF", "Depressed": "#E07B7B"},
            ax=axes[1], linewidth=1.5)
axes[1].set_title("Age Boxplot by Depression Status", fontweight="bold")
axes[1].set_xlabel("")
axes[1].set_ylabel("Age")

plt.tight_layout()
plt.savefig("viz2_age_distribution.png", bbox_inches="tight")
plt.show()
```



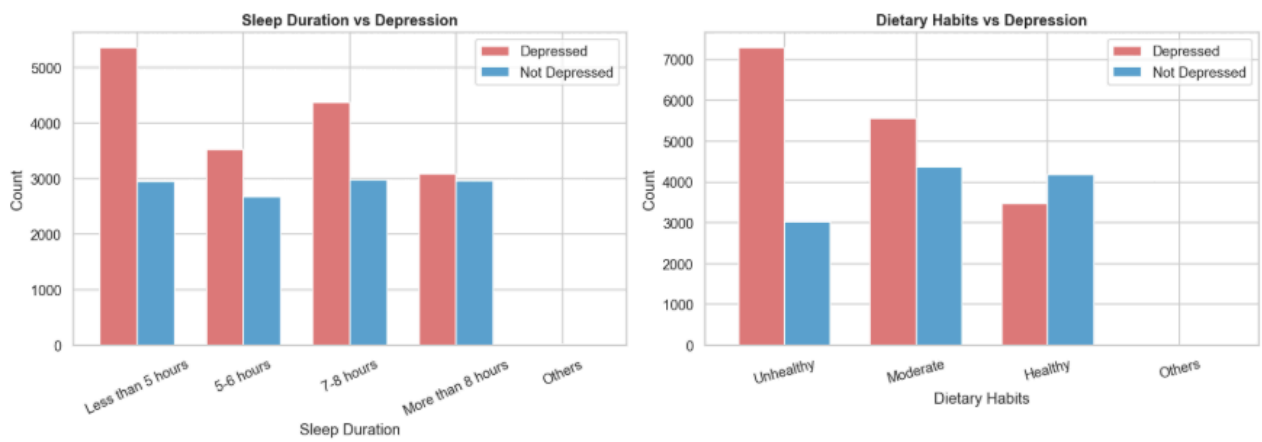
5.3 Academic Pressure and CGPA

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# --- Left: How depression changes with academic pressure level ---
sns.countplot(data=df, x="Academic Pressure", hue="Depression_Label",
              palette={"Not Depressed": "#5BA4CF", "Depressed": "#E07B7B"},
              ax=axes[0], edgecolor="white")
axes[0].set_title("Academic Pressure vs Depression", fontweight="bold")
axes[0].set_xlabel("Academic Pressure (1=Low, 5=High)")
axes[0].set_ylabel("Count")
axes[0].legend(title="Status")

# --- Right: CGPA distribution for depressed vs not ---
for val, label, color in [(0, "Not Depressed", "#5BA4CF"), (1, "Depressed", "#E07B7B")]:
    subset = df[df["Depression"] == val]["CGPA"]
    axes[1].hist(subset, bins=25, alpha=0.65, label=label, color=color, edgecolor="white")

axes[1].set_title("CGPA Distribution by Depression Status", fontweight="bold")
axes[1].set_xlabel("CGPA")
axes[1].set_ylabel("Frequency")
axes[1].legend()

plt.tight_layout()
plt.savefig("viz3_academic.png", bbox_inches="tight")
plt.show()
```



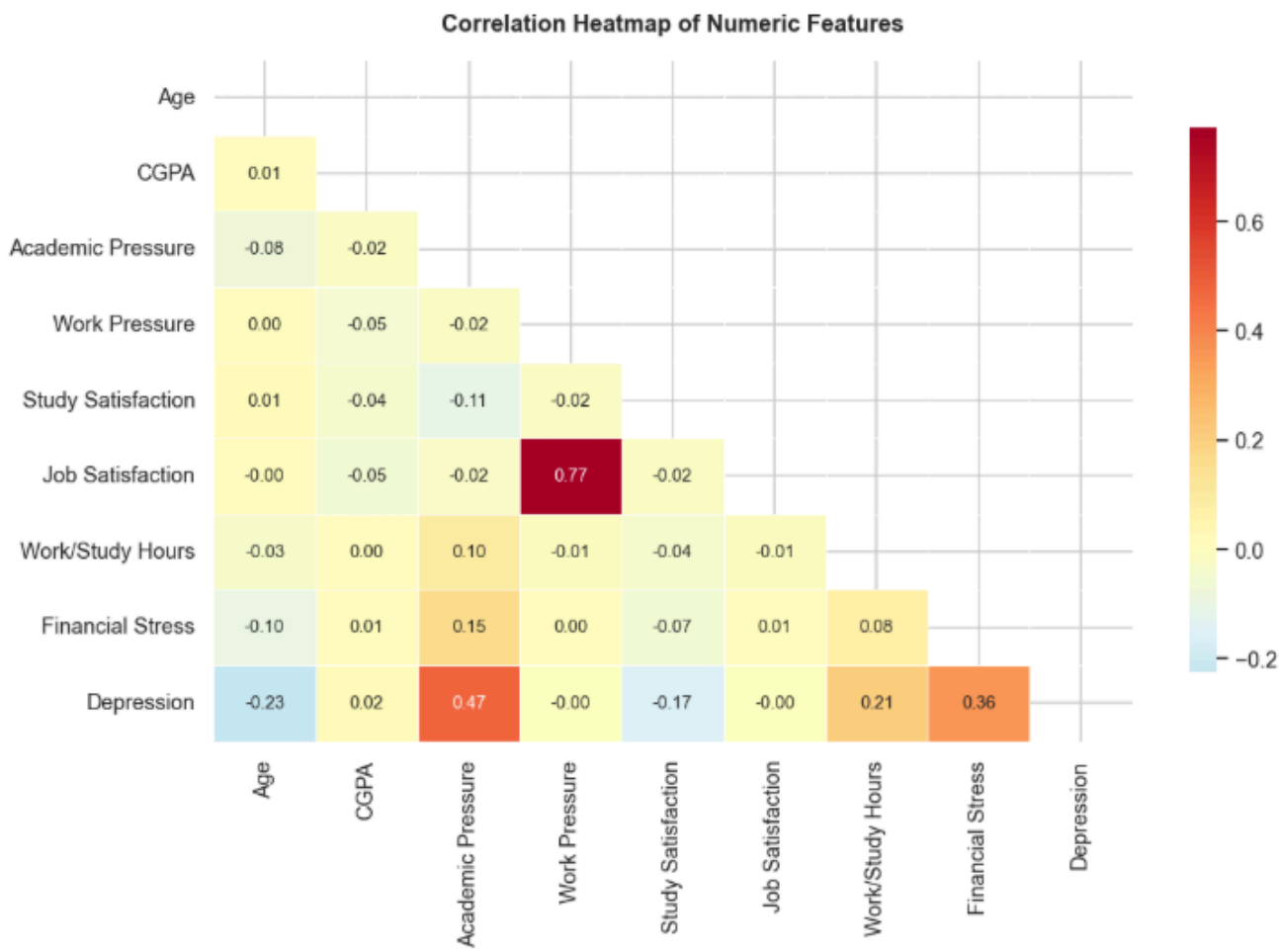
5.4 Sleep Duration and Dietary Habits

```
# Define the order we want to display these categories
sleep_order = ["Less than 5 hours", "5-6 hours", "7-8 hours", "More than 8 hours", "Others"]
diet_order  = ["Unhealthy", "Moderate", "Healthy", "Others"]

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# --- Left: Sleep duration vs depression ---
# Group and count, then reshape for easy plotting
sleep_data = df.groupby(["Sleep Duration", "Depression_Label"]).size().unstack(fill_value=0)
sleep_data = sleep_data.reindex([s for s in sleep_order if s in sleep_data.index])
sleep_data.plot(kind="bar", ax=axes[0], color=["#E07B7B", "#5BA4CF"], edgecolor="white", width=0.7)
axes[0].set_title("Sleep Duration vs Depression", fontweight="bold")
axes[0].set_xlabel("Sleep Duration")
axes[0].set_ylabel("Count")
axes[0].tick_params(axis="x", rotation=25)
axes[0].legend()

# --- Right: Dietary habits vs depression ---
diet_data = df.groupby(["Dietary Habits", "Depression_Label"]).size().unstack(fill_value=0)
diet_data = diet_data.reindex([d for d in diet_order if d in diet_data.index])
diet_data.plot(kind="bar", ax=axes[1], color=["#E07B7B", "#5BA4CF"], edgecolor="white", width=0.7)
axes[1].set_title("Dietary Habits vs Depression", fontweight="bold")
axes[1].set_xlabel("Dietary Habits")
axes[1].set_ylabel("Count")
axes[1].tick_params(axis="x", rotation=15)
axes[1].legend()

plt.tight_layout()
plt.savefig("viz4_sleep_diet.png", bbox_inches="tight")
plt.show()
```



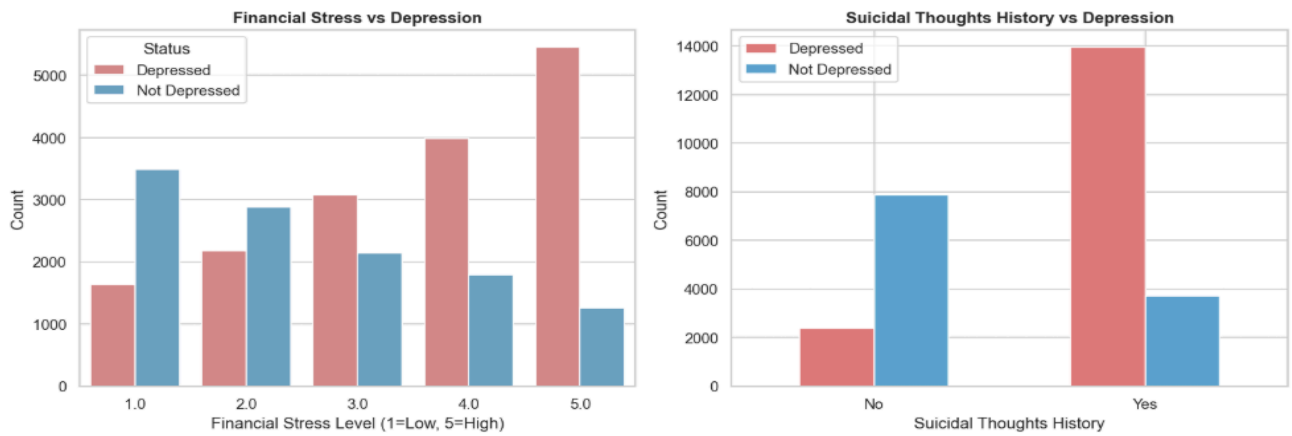
5.5 Correlation Heatmap of Numeric Features

```
# Pick only the number columns for correlation
num_cols = ["Age", "CGPA", "Academic Pressure", "Work Pressure",
            "Study Satisfaction", "Job Satisfaction",
            "Work/Study Hours", "Financial Stress", "Depression"]

# Calculate how related each pair of columns is (-1 to 1)
corr = df[num_cols].corr()

# Only show the bottom half of the table (the top half is the same)
mask = np.triu(np.ones_like(corr, dtype=bool))

fig, ax = plt.subplots(figsize=(10, 7))
sns.heatmap(corr, mask=mask, annot=True, fmt=".2f",
            cmap="RdYlBu_r", center=0, linewidths=0.5,
            ax=ax, annot_kws={"size": 9}, cbar_kws={"shrink": 0.8})
ax.set_title("Correlation Heatmap of Numeric Features", fontweight="bold", pad=15)
plt.tight_layout()
plt.savefig("viz5_heatmap.png", bbox_inches="tight")
plt.show()
```



5.6 Financial Stress and Suicidal Thoughts

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# --- Left: Financial stress level vs depression ---
sns.countplot(data=df, x="Financial Stress", hue="Depression_Label",
              palette={"Not Depressed": "#5BA4CF", "Depressed": "#E07B7B"},
              ax=axes[0], edgecolor="white")
axes[0].set_title("Financial Stress vs Depression", fontweight="bold")
axes[0].set_xlabel("Financial Stress Level (1=Low, 5=High)")
axes[0].set_ylabel("Count")
axes[0].legend(title="Status")

# --- Right: Suicidal thoughts history vs depression ---
suicidal_data = df.groupby(["Have you ever had suicidal thoughts ?", "Depression_Label"]).size().unstack(fill_value=0)
suicidal_data.plot(kind="bar", ax=axes[1], color=["#E07B7B", "#5BA4CF"], edgecolor="white", width=0.55)
axes[1].set_title("Suicidal Thoughts History vs Depression", fontweight="bold")
axes[1].set_xlabel("Suicidal Thoughts History")
axes[1].set_ylabel("Count")
axes[1].tick_params(axis="x", rotation=0)
axes[1].legend()

plt.tight_layout()
plt.savefig("viz6_stress_suicidal.png", bbox_inches="tight")
plt.show()
```



6. Observations and Interpretation

After looking at the data, here are five things I noticed:

1. More students are depressed than not. Out of 27,901 students in my dataset, about 58.5% are depressed and 41.5% are not. This means the data is a little unbalanced, with more depressed students than non-depressed ones. This is something to keep in mind when building a prediction model later. I might also trim the data a bit so that it will be balanced.

2. Higher academic pressure is linked to more depression. Students who rated their academic pressure as 4 or 5 (out of 5) were more likely to be depressed. This makes sense because the more stressed a student is about school, the harder it is to stay mentally healthy.

3. Students who sleep less tend to be more depressed. Students who sleep less than 5 hours a night made up the biggest group of depressed students. On the other hand, students who sleep 7 to 8 hours tend to have better mental health. Getting enough sleep seems to really matter.

4. Eating habits and money problems also play a role. Students with unhealthy diets had higher depression rates compared to those who eat well. Also, students with higher financial stress (level 4 or 5) were more often depressed. This shows that everyday problems like money and food choices can affect mental health too.

5. Academic pressure and financial stress are the strongest warning signs. Looking at the heatmap, academic pressure and financial stress had the highest connection to depression among all the numeric columns. Interestingly, CGPA (grades) had a very weak connection, meaning getting good or bad grades alone does not really predict if a student is depressed.